A Falsifiable, Multi-Paradigm Pipeline for Bioacoustic Structure Discovery

Sheel Morjaria

Abstract

Discovering structure in animal vocalizations without anthropomorphic contamination remains a

1. Introduction

Animal vocalizations carry rich acoustic structure. Songs, calls, and bouts exhibit recurrenc

We address these challenges with a pipeline built on three design principles:

1. **Schema-first falsifiability.** Every output is a six-axis Evidence Profile validated by a sch

2. **Ethological firewall.** Every model in the pipeline carries machine-readable

provena

3. **Semantic ceiling.** Acoustic tokens (VQ-VAE or k-means clusters) are

acoustic

Contributions

This work makes the following contributions:

A **segmentation router** that selects the appropriate segmenter per acoustic regime

(PCEN+C

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2. Related Work

Spoken-term discovery (Park & Glass, 2008; Jansen et al., 2013) finds recurring

acoustic p

Bioacoustic structure analysis has typically employed either hand-crafted features

(spectral,

Network analysis of vocal sequences (Sasahara et al., 2012; Kershenbaum et al., 2014)

models ar

Recurrence quantification analysis (Webber & Zbilut, 1994; Marwan et al., 2007) quantifies

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3. Framework

3.1 The Segmentation Router

Different acoustic regimes require fundamentally different segmenters. A single-segmenter approach

The router maintains three regimes in `configs/segmentation_router.yaml`:

Regime	Method	Strength	Example
Discrete calls in noise	PCEN+CRNN → HECB annotation	Isolates calls from noise floor	MeerKAT (8 kHz)
Continuous speech-like	NBD + CPC calibrator (ABSTAIN)	Precision via learned context	LibriSpeech (16 kHz)
Exploratory / unknown	Fixed-threshold NBD + ABSTAIN	Conservative; no false structure	Novel species

The router decision is auditable: every run produces `trace["router_audit"]` recording the detected

3.2 HECB-Seg: Hierarchical Event-Conditioned Boundary Segmentation

For calls-in-noise, the pipeline uses a two-stage architecture:

Stage 1 — Call isolation: PCEN spectrogram → Conv2D→BiLSTM → P(call) per frame → hysteresis

Stage 2 — Internal annotation: NBD proposals inside each call event → typed classifier (onset/off

A parent-quality gate (`min_parent_confidence`) abstains on low-confidence parent events, preventin

3.3 DUSTED: DTW Recurrent Acoustic Terms

DUSTED adapts spoken-term discovery to animal vocalizations. Each bout or call is represent

Validated on three structural regimes (uniform DTW threshold thr=0.15, HAC clustering):

Corpus	Regime	# Families	Coverage	Null ratio	Bootstrap
Bengalese finch	Discrete, diverse	457	0.480	**2.67×**	0.509

Egyptian fruit bat	Graded, continuous	306	0.153	1.36×	0.394
Infant marmoset	Sparse, degenerate	139	0.028	1.55×	0.229

3.4 The Acoustic Fingerprint (44-D, All-Species)

The pipeline's cross-species representation is a physics-grounded, species-portable

per-segm

Physics (~16D): spectral centroid, bandwidth, rolloff, flatness, flux, zero-crossing

rate, RM

Voiced/harmonic features are flagged and zeroed for non-tonal segments (anti-anthropomorphism).

Mel-MFCC

3.5 Ethological Firewall

Every model in the pipeline carries provenance metadata in `configs/model_metadata.json`:

pretrainin

1. Text-supervised models (NatureLM) are retrieval-only.

2. Cross-s

3.6 The Six-Axis Evidence Profile

The verdict is six **independent** axes, each categorical with a calibrated confidence:

`sequence_recombination · functional_aliasing · call_unitization · context_binding ·

turn_struc

Scorers take primitive signals (LZ complexity, alias density, Cramér's V, etc.) so the

profile is u

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4. Results

4.1 What Acoustics Encode: Individual Identity and Call Form

Corpus	Target	Metric	Result	Lift
Egyptian fruit bat	Emitter identity	Cramér's V	0.446	—
Rook	Source (individual) identity	Cramér's V	0.31	—
Marmoset	Call type	Cramér's V / linear probe	0.599 / 0.829	3.0× / 4.1×
MeerKAT	Call type	Cramér's V / linear probe	0.521 / 0.753	—
Bengalese finch	Syllable (GT)	DTW purity (wmean)	0.788	—

Acoustic structure reliably carries **individual identity** (bat, rook) and **call/event

form** (m

4.2 What Acoustics Do Not Encode: Context, Valence, and Meaning

Finding	Evidence	Section
Context requires social covariates	Bat: D(acoustic+social) = 4.3× ≫ A(acous	§9
Valence is locked in call type	Soundwel: within-call-type fp→valence 0.	§24
Social partner ≈ chance	RookID: cross-caller retrieval below cha	§26
Tokens ≠ words	LibriSpeech: token↔word macro consistenc	§29
Degenerate repertoire (token-free)	Marmoset MP: 87.5% stutter-dominant, u	§34

The **semantic ceiling** holds across all species: acoustics alone cannot recover context,

emotiona

4.3 Multi-Paradigm Structural Corroboration

To demonstrate that DUSTED's findings are not method-dependent, four independent

mathema

Paradigm	Mathematics	Dataset	Key Result	Verdict
PST + Transition Networ	Variable-order Markov + graph topology	Bengalese finch (discrete)	247 predictive contexts, 5 communities	**Corroborates.** DUSTED's discrete-
RQA	Recurrence plot → %DET, %LAM	Bat (AM)	%LAM context-dependent (Kruskal-Wallis p	**Complementary.** DUSTED where DT
Matrix Profile	STUMP continuous subsequence matching	Marmoset (sparse)	87.5% stutter-dominant, dominant motif	**Confirms.** Degenerate repertoire t
Transition Network	Networkx community detection	LibriSpeech (diverse)	639 word nodes, 19 communities, Louvain	**Universal.** Scales to human-s

Three paradigms (alignment, graph, dynamics) agree on the biological reality of the three

repertoire

4.4 Negative Controls: Pipeline Specificity

To counter the reviewer argument that "the pipeline finds structure everywhere," three

negative c

Control	Description	# Motifs	vs Natural (33)
Synthetic FM chirps	Non-biological FM-sweep sequences (250 k	**1**	33× specificity
Random audio segments	Random chunks from bat recordings (not c	**1**	33× specificity
Order-shuffled bat	Same units, permuted order	22	Partial collapse

The pipeline finds **1 motif** in synthetic and random audio versus **33** in natural bat

bouts—a

4.5 Boundary-Quality Sweep: Whole-Call Primary

The boundary-quality sensitivity sweep varies parent-event boundary quality (oracle GT →

CRNN at r

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5. The H0-CMC Evidence Ladder

The Causal Multimodal Convergence (H0-CMC) framework defines a six-rung evidence ladder

connectin

Tier	Evidence	Autonomous?	Semantic validation
T0 — Acoustic	Recurrent acoustic structure found		none

T1 — Context	Context-associated cue (binding signific		none (correlation)
T2 — Playback	Counterfactual stimuli generated, respond	Structurally	none (response pending)
T3 — Convergence	Multimodal convergence (acoustic + behav	Within-lab	candidate_functional_interpretation
T4 — Counterfactual	Counterfactual disrupts receiver response	Within-lab	causally_supported_signal
T5 — Field	Ecologically validated (fitness-relevant	Natural	ecologically_validated_function

The pipeline operates autonomously at T0–T1. It generates T2 playback stimuli via the DSP Audio

An explicit-implicit test classifies each corpus's communication mode: if acoustics alone recover co

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6. Track C: Playback Readiness

The DSP Audio Splicer (`experiments/playback_design/audio_splicer.py`) generates falsifiable

Condition	Transform	Hypothesis	H0-CMC tier
Natural	Unchanged baseline	—	—
Order shuffled	Permute unit order	Does sequence syntax matter?	T2
Onset–offset swap	Swap first and last units	Do positional roles matter?	T2
Caller swapped	Replace one unit with a donor from another	Does caller identity matter?	T2
Amplitude matched	RMS-normalize to target	Control: rules out loudness confound	—

Stimuli are generated via exemplar-based DSP (slice + crossfade + concatenate), not neural synthesis.

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7. Discussion

7.1 The Three-Bucket Thesis

Every finding in this pipeline lands in exactly one of three buckets:

1. **What acoustics encode.** Individual identity (bat Emitter V = 0.446, rook Source V = 0.31) ar

2. **What acoustics do not encode.** Context (bat requires social covariates; social_lift = 1.59)

3. **What requires social or biological validation.** All functional interpretation. The H0-CMC

7.2 Honest Limitations

Tokenization. DUSTED operates on k-means cluster IDs from the acoustic fingerprint, not learn

7.3 What This Pipeline Is Not

This pipeline is explicitly **not a classifier** that emits labels. It does not claim validated

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8. Conclusion

We have presented a falsifiable, multi-paradigm pipeline for bioacoustic structure discovery

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Reproducibility

All code, tests (247), and validated results (35 benchmark sections) are in the `bioacous

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